

第七章 哈密顿力学

§0 勒让德变换

$$F = F(w, u)$$

$$dF = \frac{\partial F}{\partial w} dw + \frac{\partial F}{\partial u} du$$

$$\text{令 } v = \frac{\partial F}{\partial u} \rightarrow dF = \frac{\partial F}{\partial w} dw + v du$$

$$\text{定义 } G = uv - F = G(w, v)$$

$$dG = \frac{\partial G}{\partial w} dw + \frac{\partial G}{\partial v} dv$$

$$= d(uv - F) = u dv + v du - dF$$

$$= - \frac{\partial F}{\partial w} dw + u dv$$

$$\Rightarrow \begin{cases} \frac{\partial G}{\partial w} = - \frac{\partial F}{\partial w} & \text{是对偶变换} \\ \frac{\partial G}{\partial v} = u \end{cases}$$

几何意义:

$$y = f(x), \quad \text{设 } k(x) := \frac{dy}{dx}(x)$$

$$dy = k dx = d(kx) - x dk$$

$$\Rightarrow \text{令 } g(k) = kx - f(x), \quad dg(k) = x dk$$

考虑 $f(x)$ 在 x_0 处的切线:

$$y = kx + (y_0 - kx_0)$$

$$\hookrightarrow g(k) = kx_0 - y_0$$

给定 k , $g(k)$ 给出截距, 对应一条原函数的切线

§ 7.1 哈密顿正则方程

拉氏力学: $q_\alpha, \dot{q}_\alpha$

哈氏力学: q_α, p_α

量纲上: $[q_\alpha \dot{q}_\alpha] = [\frac{q_\alpha^2}{t}]$

$$[p_\alpha q_\alpha] = [\frac{\partial L}{\partial \dot{q}_\alpha} q_\alpha] = [L t] \text{ "作用量" 与 } q_\alpha \text{ 量纲无关}$$

(★ 拉格朗日方程: $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial q_\alpha} = 0 \quad (\alpha = 1, \dots, s)$)

1. 定义广义动量 $p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$

$$\text{则 } \star \rightarrow p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$$

$$\text{或用便 } (q_\alpha, \dot{q}_\alpha) \rightarrow (q_\alpha, p_\alpha)$$

$$\text{即 } L(q, \dot{q}, t) \rightarrow L(q, \underline{q(p)}, t)$$

$$\underline{q(q, p)} \text{ 可由 } p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha} \text{ 解出}$$

$$\bar{L}(q, p, t) = L[q, \underline{q(q, p, t)}, t]$$

$$\begin{cases} \frac{\partial \bar{L}}{\partial q_\alpha} = \frac{\partial L}{\partial q_\alpha} + \sum_{\beta} \frac{\partial L}{\partial \dot{q}_\beta} \frac{\partial \dot{q}_\beta}{\partial q_\alpha} \\ \frac{\partial \bar{L}}{\partial p_\alpha} = \sum_{\beta} \frac{\partial L}{\partial \dot{q}_\beta} \frac{\partial \dot{q}_\beta}{\partial p_\alpha} \end{cases}$$

$$\rightarrow \begin{cases} \frac{\partial \bar{L}}{\partial q_\alpha} = p_\alpha + \sum_{\beta} p_\beta \frac{\partial \dot{q}_\beta}{\partial q_\alpha} \\ \frac{\partial \bar{L}}{\partial p_\alpha} = \sum_{\beta} p_\beta \frac{\partial \dot{q}_\beta}{\partial p_\alpha} \end{cases} \text{ 稍简化}$$

注意 $\sum_{\beta} p_\beta \frac{\partial \dot{q}_\beta}{\partial q_\alpha} = \sum_{\beta} \frac{\partial}{\partial q_\alpha} (p_\beta \dot{q}_\beta)$, $\sum_{\beta} p_\beta \frac{\partial \dot{q}_\beta}{\partial p_\alpha} = \sum_{\beta} \frac{\partial}{\partial p_\alpha} (p_\beta \dot{q}_\beta) - \dot{q}_\alpha$
 到 "勒让德变换" $\frac{\partial \bar{L}}{\partial q} = 0$

$$\rightarrow \begin{cases} \frac{\partial \bar{L}}{\partial q_\alpha} = \dot{p}_\alpha + \sum_\beta \frac{\partial}{\partial q_\alpha} (p_\beta q_\beta) \\ \frac{\partial \bar{L}}{\partial p_\alpha} = \sum_\beta \frac{\partial}{\partial p_\alpha} (p_\beta q_\beta) - q_\alpha \end{cases}$$

$$\rightarrow \begin{cases} \sum_\beta \frac{\partial}{\partial q_\alpha} (p_\beta q_\beta - \bar{L}) = -\dot{p}_\alpha \\ \sum_\beta \frac{\partial}{\partial p_\alpha} (p_\beta q_\beta - \bar{L}) = q_\alpha \end{cases}$$

定义哈密顿函数 $H = p_\alpha q_\alpha - \bar{L}$

$\rightarrow$ 哈密顿正则方程

形式简单而对称

$p_\alpha, q_\alpha \rightarrow$ 正则变量

$$\begin{cases} \frac{\partial H}{\partial q_\alpha} = -\dot{p}_\alpha \\ \frac{\partial H}{\partial p_\alpha} = q_\alpha \end{cases}$$

$$2. \quad dL(q, \dot{q}, t) = \sum_\alpha \frac{\partial L}{\partial \dot{q}_\alpha} d\dot{q}_\alpha + \sum_\alpha \frac{\partial L}{\partial q_\alpha} dq_\alpha + \frac{\partial L}{\partial t} dt$$

代入 $p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}$ (广义动量定义), $\dot{p}_\alpha = \frac{\partial L}{\partial q_\alpha}$ (拉格朗日方程)

$$dL(q, \dot{q}, t) = \sum_\alpha \dot{p}_\alpha dq_\alpha + \sum_\alpha p_\alpha d\underline{q}_\alpha + \frac{\partial L}{\partial t} dt$$

结果为 dp_α

$$= d\bar{L}(q, p, t)$$

勒让德变换:

$$d\left(\sum_\alpha p_\alpha q_\alpha\right) = \sum_\alpha (q_\alpha dp_\alpha + p_\alpha dq_\alpha)$$

$$dL = d\bar{L} = \sum_\alpha \dot{p}_\alpha dq_\alpha + d\left(\sum_\alpha p_\alpha q_\alpha\right) - \sum_\alpha q_\alpha dp_\alpha + \frac{\partial L}{\partial t} dt$$

$$d\left(\sum_\alpha p_\alpha q_\alpha - L\right) = \sum_\alpha q_\alpha dp_\alpha - \sum_\alpha \dot{p}_\alpha dq_\alpha - \frac{\partial L}{\partial t} dt$$

$$\dot{H} \propto H = \sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} - L$$

$$\begin{aligned} dH(q, p, t) &= \sum_{\alpha} \frac{\partial H}{\partial \dot{q}_{\alpha}} d\dot{q}_{\alpha} + \sum_{\alpha} \frac{\partial H}{\partial p_{\alpha}} dp_{\alpha} + \frac{\partial H}{\partial t} dt \\ &\stackrel{\text{对 } \dot{q}}{\downarrow} = \sum_{\alpha} \underline{(-p_{\alpha})} d\dot{q}_{\alpha} + \sum_{\alpha} \underline{\dot{q}_{\alpha}} dp_{\alpha} - \frac{\partial L}{\partial t} dt \end{aligned}$$

$$\Rightarrow \begin{cases} \frac{\partial H}{\partial \dot{q}_{\alpha}} = -p_{\alpha} \\ \frac{\partial H}{\partial p_{\alpha}} = \dot{q}_{\alpha} \\ \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \end{cases}$$

3. 运动积分

1) 广义动量积分

$$\frac{\partial H}{\partial \dot{q}_{\alpha}} = 0 = -p_{\alpha} \quad \Rightarrow \quad p_{\alpha} = \text{const.}$$

$q_{\alpha} \rightarrow$ 可造 / 循环坐标

$$\frac{\partial H}{\partial q_{\alpha}} = \frac{\partial}{\partial q_{\alpha}} \left(\sum_{\alpha} p_{\alpha} \dot{q}_{\alpha}(q, p, t) - L(q, \dot{q}(q, p, t), t) \right)$$

$$= \sum_{\alpha} \left[p_{\alpha} \frac{\partial \dot{q}_{\alpha}}{\partial q_{\alpha}} - \left(\frac{\partial L}{\partial q_{\alpha}} + \underbrace{\frac{\partial L}{\partial \dot{q}_{\alpha}} \frac{\partial \dot{q}_{\alpha}}{\partial q_{\alpha}}}_{= p_{\alpha}} \right) \right]$$

$$= - \frac{\partial L}{\partial q_{\alpha}} \quad \frac{\partial H}{\partial q_{\alpha}} = - \frac{\partial L}{\partial q_{\alpha}} \quad \text{隐含在勒让德变换中?}$$

$\Rightarrow$ 对 $\dot{q}$ $L \rightarrow H$ 的变换, 可造坐标不变

2) 广义能量积分

$$\frac{dH}{dt} = \sum_{\alpha} \frac{\partial H}{\partial \dot{q}_{\alpha}} \cdot \dot{q}_{\alpha} + \sum_{\alpha} \frac{\partial H}{\partial p_{\alpha}} \cdot \dot{p}_{\alpha} + \frac{\partial H}{\partial t}$$

$$= \sum_{\alpha} [-p_{\alpha} \dot{q}_{\alpha} + \dot{q}_{\alpha} p_{\alpha}] + \frac{dH}{dt} = \frac{dH}{dt}$$

$\Rightarrow \frac{dH}{dt} = - \frac{\partial L}{\partial t}$ 在 $r_i = r_i(q_1, \dots, q_s)$ 的变换下, H 为机械能

例:

1. 弹簧振子

$$L = T - V = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2 \quad p = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$H = p \dot{x} - L = \frac{p^2}{2m} + \frac{1}{2} k x^2$$

$$\begin{cases} \frac{\partial H}{\partial q_i} = -\dot{p}_i \\ \frac{\partial H}{\partial p_i} = \dot{q}_i \end{cases} \rightarrow \begin{cases} \frac{\partial H}{\partial x} = kx = -\dot{p} \\ \frac{\partial H}{\partial p} = \frac{p}{m} = \dot{x} \quad (\dot{p} = m\ddot{x}) \end{cases}$$

$$\rightarrow m\ddot{x} = -kx$$

$$\text{解为 } x = A \cos(\sqrt{\frac{k}{m}}t + \varphi)$$

$$p = m\dot{x} = -A \cdot \sqrt{km} \sin(\sqrt{\frac{k}{m}}t + \varphi)$$

$$E = H = \frac{1}{2} k x^2 + \frac{p^2}{2m} \rightarrow A = \sqrt{\frac{2E}{k}}$$

$$\rightarrow \begin{cases} x = \sqrt{\frac{2E}{k}} \cos(\sqrt{\frac{k}{m}}t + \varphi) \\ p = -\sqrt{2Em} \sin(\sqrt{\frac{k}{m}}t + \varphi) \end{cases}$$

2. Atwood 机

3.

4.

5.

§ 7.2 相空间 刘维尔定理

$(X, P) \rightarrow$ "相"

相空间 & 相轨道

一般地, 相 $(q_1 \dots q_s, p_1 \dots p_s)$

相空间为 $2s$ 维的

多自由度 $\rightarrow$ 取出 "代表点"

系综: 性质相同的力学系统集合

刘维尔定理: 代表点在空间中运动时, 密度 ρ 不变

$$\text{定义相空间中体积元 } dV = \prod_{i=1}^s dq_i dp_i$$

代表点个数 N

$$\text{定义密度 } \rho = \frac{dN}{dV} = \rho(q, p, t) \quad (\text{充分大又充分小})$$

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i} \dot{q}_i + \frac{\partial \rho}{\partial p_i} \dot{p}_i \right)$$

$$\frac{\partial \rho}{\partial t}: \text{对于一相空间 } q_i, q_i + dq_i$$

$$dV = dA_1 dq_i, \quad (A_1 \text{ 为 } i \text{ 义面积 } dA_1 = \prod_{j \neq i} dq_j \prod_{k=1}^s dp_k)$$

at 时间由:

$$\text{进入点数 } dN_{in, q_i} = \rho(q_i) dA_1|_{q_i} \cdot v_i \cdot dt$$

$$\text{离开点数 } dN_{out, q_i} = \rho(q_i) dA_1|_{q_i + dq_i} \cdot v_i \cdot dt$$

$$\rightarrow \frac{dN_{in}}{dt} = dA_1 (\rho(q_i)|_{q_i} - \rho(q_i)|_{q_i + dq_i})$$

$$= - \frac{p}{\partial q_1} dq_1 \cdot dA_1 = - \frac{p}{\partial q_1} dV$$

带来 p 变化:

$$\frac{\partial p_{q_1}}{\partial t} = \frac{\partial}{\partial t} \frac{dN_1}{dV} = \frac{1}{dV} \frac{\partial N_1}{\partial t} = - \frac{\partial (p q_1)}{\partial q_1}$$

$$\frac{\partial p}{\partial t} = \sum_i \frac{\partial p_i}{\partial t} = \sum_\alpha \left(\frac{\partial (p q_\alpha)}{\partial q_\alpha} + \frac{\partial (p \dot{p}_\alpha)}{\partial p_\alpha} \right)$$

同理:

$$\frac{dp}{dt} = \sum_\alpha \left(\frac{\partial (p q_\alpha)}{\partial q_\alpha} + \frac{\partial (p \dot{p}_\alpha)}{\partial p_\alpha} \right) + \sum_\alpha \left(\frac{dp}{dq_\alpha} q_\alpha + \frac{dp}{dp_\alpha} \dot{p}_\alpha \right)$$

$$= - \sum_\alpha p \left(\frac{\partial q_\alpha}{\partial q_\alpha} + \frac{\partial \dot{p}_\alpha}{\partial p_\alpha} \right) \quad q_\alpha = \frac{\partial H}{\partial p_\alpha} \quad \dot{p}_\alpha = - \frac{\partial H}{\partial q_\alpha}$$

$$= - \sum_\alpha p \left[\frac{\partial^2 H}{\partial q_\alpha \partial p_\alpha} + \left(- \frac{\partial^2 H}{\partial p_\alpha \partial q_\alpha} \right) \right]$$

$$= 0$$

§ 7.3 位力定理

$$\text{证} \times \quad S = \sum_i \vec{p}_i \cdot \vec{r}_i$$

$$\frac{dS}{dt} = \sum_i (\dot{\vec{p}}_i \cdot \vec{r}_i + \vec{p}_i \cdot \dot{\vec{r}}_i) = \sum_i (\vec{F}_i \cdot \vec{r}_i + m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i)$$

$$= \sum_i \vec{F}_i \cdot \vec{r}_i + 2T$$

取平均

$$\left\langle \frac{dS}{dt} \right\rangle = \frac{1}{\tau} \int_0^\tau \dot{S} dt = \frac{S(\tau) - S(0)}{\tau}$$

$$\tau \rightarrow \infty \quad \text{而 } S \text{ 有界} \Rightarrow \left\langle \frac{dS}{dt} \right\rangle \rightarrow 0$$

$$0 = \langle \sum \vec{F}_i \cdot \vec{r}_i \rangle + \underbrace{\langle \sum \vec{F}_i \cdot \vec{r}_i \rangle}_{\text{位力}}$$

考虑仅受保守力的系统

$$\vec{F}_i = -\nabla V_i \quad \text{即} \quad \langle T \rangle = \left\langle \frac{1}{2} \sum_i \vec{F}_i \cdot \nabla V_i \right\rangle$$

考虑万有引力下的 $V = -\frac{k}{r} \quad \nabla V = \frac{k}{r^2}$

$$\langle T \rangle = \left\langle \frac{1}{2} \frac{k}{r} \right\rangle = -\frac{1}{2} \langle V \rangle$$

例:

例: 在二体问题的椭圆轨道情形验证位力定理

例: 用位力定理推导理想气体的状态方程

$$\langle T \rangle = \left\langle \sum_i \frac{1}{2} m_i v_i^2 \right\rangle = \frac{3}{2} n k T$$

$$\left\langle \sum_i \vec{F}_i \cdot \vec{r}_i \right\rangle = \int_S (-p d\vec{S}) \cdot \vec{r}$$

Gauss 定理: $\int_S \vec{A} \cdot d\vec{S} = \int_V (\nabla \cdot \vec{A}) dV$

$$\int_S (-p d\vec{S}) \cdot \vec{r} = - \int_V (\nabla \cdot p \vec{r}) dV$$

$$[\vec{r}] = x \cdot \hat{x} + y \cdot \hat{y} + z \cdot \hat{z}$$

$$= -p \int_V \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} \right) dV = -3pV$$

$$3 n k T + (-3pV) = 0 \quad \Rightarrow \quad pV = n k T = \mu R T$$

§ 7.4 泊松括号

1.

考虑力学量 $\varphi(q, p, t)$

$$\begin{aligned}\frac{d\varphi}{dt} &= \frac{\partial \varphi}{\partial t} + \sum_{\alpha} \left(\frac{\partial \varphi}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial \varphi}{\partial p_{\alpha}} \dot{p}_{\alpha} \right) \\ &= \frac{\partial \varphi}{\partial t} + \sum_{\alpha} \left(\frac{\partial \varphi}{\partial q_{\alpha}} \cdot \frac{\partial H}{\partial p_{\alpha}} - \frac{\partial \varphi}{\partial p_{\alpha}} \cdot \frac{\partial H}{\partial q_{\alpha}} \right)\end{aligned}$$

2.

定义 泊松括号 $[\varphi, \psi]$

$$[\varphi, \psi] = \sum_{\alpha} \left(\frac{\partial \varphi}{\partial q_{\alpha}} \cdot \frac{\partial \psi}{\partial p_{\alpha}} - \frac{\partial \varphi}{\partial p_{\alpha}} \cdot \frac{\partial \psi}{\partial q_{\alpha}} \right)$$

$$\rightarrow \frac{d\varphi}{dt} = \frac{\partial \varphi}{\partial t} + [\varphi, H]$$

哈密顿方程

$$\begin{cases} \dot{p}_{\alpha} = - \frac{\partial H}{\partial q_{\alpha}} = \sum_{\beta} \left(\frac{\partial p_{\alpha}}{\partial q_{\beta}} \cdot \frac{\partial H}{\partial p_{\beta}} - \frac{\partial p_{\alpha}}{\partial p_{\beta}} \cdot \frac{\partial H}{\partial q_{\beta}} \right) = [p_{\alpha}, H] \\ \dot{q}_{\alpha} = \frac{\partial H}{\partial p_{\alpha}} = \sum_{\beta} \left(\frac{\partial q_{\alpha}}{\partial q_{\beta}} \cdot \frac{\partial H}{\partial p_{\beta}} - \frac{\partial q_{\alpha}}{\partial p_{\beta}} \cdot \frac{\partial H}{\partial q_{\beta}} \right) = [q_{\alpha}, H] \end{cases}$$

$$\textcircled{1} [q_{\alpha}, q_{\beta}] = 0 \quad \textcircled{2} [p_{\alpha}, p_{\beta}] = 0 \quad \textcircled{3} [q_{\alpha}, p_{\beta}] = \delta_{\alpha\beta}$$

$$[\varphi, c] = 0, [\varphi, t] = 0 \quad [\varphi, \psi] = -[\psi, \varphi]$$

$$[\varphi, \sum_j C_j \psi_j] = \sum_j C_j [\varphi, \psi_j]$$

$$[\varphi, \psi_1 \psi_2] = [\varphi, \psi_1] \psi_2 + [\varphi, \psi_2] \psi_1$$

$$\frac{\partial}{\partial t} [\varphi, \psi] = \left[\frac{\partial \varphi}{\partial t}, \psi \right] + \left[\varphi, \frac{\partial \psi}{\partial t} \right]$$

$$[q_{\alpha}, p_{\beta}] = \delta_{\alpha\beta}$$

$$[p_\alpha, \psi] = - \frac{\partial \psi}{\partial q_\alpha}$$

3.

$$\text{验证可比较等式: } [\varphi, [\psi, \theta]] + [\psi, [\theta, \varphi]] + [\theta, [\varphi, \psi]] = 0$$

(轮换和)

$$\begin{aligned} \text{证明: } [\psi, \theta] &= \sum_{\alpha} \left(\frac{\partial \psi}{\partial q_\alpha} \frac{\partial \theta}{\partial p_\alpha} - \frac{\partial \psi}{\partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha} \right) \\ &= \sum_{\alpha} \left(\frac{\partial \psi}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} - \frac{\partial \psi}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} \right) \cdot \theta = D_1 \theta \end{aligned}$$

$$[\varphi, \theta] = \sum_{\alpha} \left(\frac{\partial \varphi}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} - \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} \right) \cdot \theta = D_2 \theta$$

$$[\varphi, [\psi, \theta]] + [\psi, [\theta, \varphi]] = -[\theta, [\varphi, \psi]]$$

$$= [\varphi, D_1 \theta] - [\psi, D_2 \theta]$$

$$= \sum_{\alpha} \left(\frac{\partial \varphi}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} (D_1 \theta) - \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} (D_1 \theta) \right) - \sum_{\alpha} \left(\frac{\partial \psi}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} (D_2 \theta) + \frac{\partial \psi}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} (D_2 \theta) \right)$$

$$= \sum_{\alpha} \frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial p_\alpha} + \underbrace{\frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial p_\alpha}}_1 - \frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha^2} \frac{\partial \theta}{\partial q_\alpha} - \underbrace{\frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_2$$

$$- \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha^2} \frac{\partial \theta}{\partial p_\alpha} - \underbrace{\frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_3 + \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha} + \underbrace{\frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_4$$

$$- \frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial p_\alpha} - \underbrace{\frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial p_\alpha}}_1 + \frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha^2} \frac{\partial \theta}{\partial q_\alpha} + \underbrace{\frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_2$$

$$+ \frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha^2} \frac{\partial \theta}{\partial p_\alpha} + \underbrace{\frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_3 - \frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha} - \underbrace{\frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial q_\alpha}}_4$$

$$= \sum_{\alpha} \frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial p_\alpha} - \frac{\partial \varphi}{\partial q_\alpha} \frac{\partial^2 \psi}{\partial p_\alpha^2} \frac{\partial \theta}{\partial q_\alpha} - \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha^2} \frac{\partial \theta}{\partial p_\alpha} + \frac{\partial \varphi}{\partial p_\alpha} \frac{\partial^2 \psi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha} \quad 4$$

$$- \frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha \partial q_\alpha} \frac{\partial \theta}{\partial p_\alpha} + \frac{\partial \psi}{\partial q_\alpha} \frac{\partial^2 \varphi}{\partial p_\alpha^2} \frac{\partial \theta}{\partial q_\alpha} + \frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha^2} \frac{\partial \theta}{\partial p_\alpha} - \frac{\partial \psi}{\partial p_\alpha} \frac{\partial^2 \varphi}{\partial q_\alpha \partial p_\alpha} \frac{\partial \theta}{\partial q_\alpha} \quad 4$$

$$- [\theta, [\varphi, \psi]] = \left[\left(\frac{\partial \theta}{\partial q_\alpha} \frac{\partial \varphi}{\partial p_\alpha} - \frac{\partial \theta}{\partial p_\alpha} \frac{\partial \varphi}{\partial q_\alpha} \right), \theta \right]$$

$$= \sum_{\alpha} \left(\frac{\partial}{\partial q_\alpha} \left(\frac{\partial \theta}{\partial q_\alpha} \frac{\partial \varphi}{\partial p_\alpha} - \frac{\partial \theta}{\partial p_\alpha} \frac{\partial \varphi}{\partial q_\alpha} \right) \frac{\partial \theta}{\partial p_\alpha} - \frac{\partial}{\partial p_\alpha} \left(\frac{\partial \theta}{\partial q_\alpha} \frac{\partial \varphi}{\partial p_\alpha} - \frac{\partial \theta}{\partial p_\alpha} \frac{\partial \varphi}{\partial q_\alpha} \right) \frac{\partial \theta}{\partial q_\alpha} \right)$$

$$= \sum \left[\frac{\partial^2 \psi}{\partial q_a^2} \frac{\partial \psi}{\partial p_a} \frac{\partial \theta}{\partial p_a} + \frac{\partial \psi}{\partial q_a} \frac{\partial^2 \psi}{\partial q_a \partial p_a} \frac{\partial \theta}{\partial p_a} - \frac{\partial^2 \psi}{\partial p_a \partial p_a} \frac{\partial \psi}{\partial q_a} \frac{\partial \theta}{\partial p_a} - \frac{\partial \psi}{\partial p_a} \frac{\partial^2 \psi}{\partial q_a^2} \frac{\partial \theta}{\partial p_a} \right. \\ \left. - \frac{\partial^2 \psi}{\partial p_a \partial q_a} \frac{\partial \psi}{\partial p_a} \frac{\partial \theta}{\partial q_a} - \frac{\partial \psi}{\partial q_a} \frac{\partial^2 \psi}{\partial p_a^2} \frac{\partial \theta}{\partial q_a} + \frac{\partial^2 \psi}{\partial p_a^2} \frac{\partial \psi}{\partial q_a} \frac{\partial \theta}{\partial q_a} + \frac{\partial \psi}{\partial p_a} \frac{\partial^2 \psi}{\partial p_a \partial q_a} \frac{\partial \theta}{\partial q_a} \right] \quad (2)$$

$$① = ② \Rightarrow [\psi, \mathcal{H}(\psi)] + [\psi, \mathcal{H}(\psi)] + [\mathcal{H}, \mathcal{H}(\psi)] = 0$$

泊松定理:

$$\text{运动积分 } \psi \text{ 满足 } \frac{d\psi}{dt} = 0$$

$$\text{即 } [\psi, H] = -\frac{\partial \psi}{\partial t}$$

$$\text{若 } \varphi, \psi \text{ 分别为运动积分 即 } [\varphi, H] = -\frac{\partial \varphi}{\partial t}, [\psi, H] = -\frac{\partial \psi}{\partial t}$$

由 Jacobi 恒等式:

$$[\varphi, [\psi, H]] + [\psi, [H, \varphi]] + [H, [\varphi, \psi]] = 0$$

$$\Rightarrow [[\varphi, \psi], H] = -[\varphi, \frac{\partial \psi}{\partial t}] + [\psi, \frac{\partial \varphi}{\partial t}] \\ = -([\varphi, \frac{\partial \psi}{\partial t}] + [\frac{\partial \varphi}{\partial t}, \psi]) \\ = -\frac{\partial}{\partial t} [\varphi, \psi]$$

$$\Rightarrow \frac{d[\varphi, \psi]}{dt} = 0 \quad \text{即 } [\varphi, \psi] \text{ 亦为运动积分}$$

推论: 若系统不显含时间, ψ 是运动积分, 则 $\frac{\partial^2 \psi}{\partial t^2}$ 也是运动积分

$$\text{证: } H \text{ 也是运动积分, } [\psi, H] = -\frac{\partial \psi}{\partial t}$$

可积系统: S 个自由度

$$\text{有 } S \text{ 个独立运动积分 } \phi_i \text{ 且有 } [\phi_a, \phi_b] = 0$$

称 $\phi_1 \sim \phi_c$ 是相互对合的 $\Rightarrow$ 刘维尔可积的

4. 量子力学

正则量子化:

$$[X, Y] \rightarrow \frac{1}{i\hbar} [\hat{X}, \hat{Y}] = \frac{1}{i\hbar} [\hat{X}\hat{Y} - \hat{Y}\hat{X}]$$

5. 一些讨论

p, q 是否独立?

① $q, \dot{q}$ 独立, $p = \frac{\partial L}{\partial \dot{q}} \Rightarrow p, q$ 独立

② 直接考虑相空间中哈密顿原理:

$$\int_t L dt = \int_t (\sum \bar{p}_\alpha \dot{q}_\alpha - H) dt = 0$$

$$= \int_t \sum (p_\alpha \delta q_\alpha + q_\alpha \delta p_\alpha - \frac{\partial H}{\partial p_\alpha} \delta p_\alpha - \frac{\partial H}{\partial q_\alpha} \delta q_\alpha) dt = 0$$

其中 $\int_t \sum p_\alpha \delta q_\alpha dt$ 在相空间内, 只需要 $\delta q_\alpha, \delta p_\alpha$

$$\text{改写为} = \sum p_\alpha \delta q_\alpha \Big|_t - \int_t \sum \dot{p}_\alpha \delta q_\alpha dt$$

在初、末状态下, $\delta q_\alpha = 0$ (坐标) 给定

$$\Rightarrow \sum p_\alpha \delta q_\alpha \Big|_t = 0 - 0 = 0$$

$$\Rightarrow \int_t \sum p_\alpha \delta q_\alpha dt = - \int_t \sum \dot{p}_\alpha \delta q_\alpha dt$$

$$\Rightarrow 0 = \int_t L dt$$

$$= \int_t \sum (p_\alpha \delta q_\alpha + q_\alpha \delta p_\alpha - \frac{\partial H}{\partial p_\alpha} \delta p_\alpha - \frac{\partial H}{\partial q_\alpha} \delta q_\alpha) dt$$

$$= \int_t \sum (-\dot{p}_\alpha \delta q_\alpha + q_\alpha \delta p_\alpha - \frac{\partial H}{\partial p_\alpha} \delta p_\alpha - \frac{\partial H}{\partial q_\alpha} \delta q_\alpha) dt$$

$$= \int_t \sum [(-\dot{p}_\alpha - \frac{\partial H}{\partial q_\alpha}) \delta q_\alpha + (q_\alpha - \frac{\partial H}{\partial p_\alpha}) \delta p_\alpha] dt = 0$$

$$\Rightarrow p_\alpha = [p_\alpha, H] \quad q_\alpha = [q_\alpha, H]$$

$$\frac{d}{dt} \frac{\partial \bar{q}_\alpha}{\partial \dot{q}_\alpha} = \frac{\partial \bar{q}_\alpha}{\partial \dot{q}_\alpha} \left[\sum_\beta \dot{P}_\beta \dot{q}_\beta (q, p, t) - H(q, p(\dot{q}, t), t) \right]$$

$$= P_\alpha + \sum_\beta \left(\frac{\partial P_\beta}{\partial \dot{q}_\alpha} \cdot \dot{q}_\beta - \frac{\partial H}{\partial P_\beta} \cdot \frac{\partial P_\beta}{\partial \dot{q}_\alpha} \right)$$

$$= P_\alpha + \sum_\beta \frac{\partial P_\beta}{\partial \dot{q}_\alpha} (\dot{q}_\beta - \frac{\partial H}{\partial P_\beta}) = P_\alpha$$

例: $H = \frac{P^2}{2m} + \frac{1}{2} k X^2$

$$a = \sqrt{\frac{m\omega}{2}} \left(X + i \frac{P}{m\omega} \right), \quad a^* = \sqrt{\frac{m\omega}{2}} \left(X - i \frac{P}{m\omega} \right) \quad \text{† 阿贝尔}$$

$$\Rightarrow H = \omega a a^\dagger$$

证: (a) $[a, a^*] = \left[\sqrt{\frac{m\omega}{2}} \left(X + i \frac{P}{m\omega} \right), \sqrt{\frac{m\omega}{2}} \left(X - i \frac{P}{m\omega} \right) \right]$

$$= \frac{m\omega}{2} \left(-\frac{i}{m\omega} - \frac{i}{m\omega} \right)$$

$$= -i$$

(b) $[a, H] = \left[\sqrt{\frac{m\omega}{2}} \left(X + i \frac{P}{m\omega} \right), H \right]$

$$= \sqrt{\frac{m\omega}{2}} \left[\frac{\partial H}{\partial P} + \frac{i}{m\omega} \left(\frac{\partial H}{\partial X} \right) \right]$$

$$= \sqrt{\frac{m\omega}{2}} \left(\dot{X} + i \frac{\dot{P}}{m\omega} \right) = \sqrt{\frac{m\omega}{2}} \left(\frac{P}{m} + i \frac{X}{\omega} \right)$$

$$= \frac{\omega}{i} \sqrt{\frac{m\omega}{2}} \left(X + i \frac{P}{m\omega} \right) = -i\omega a$$

(c) $[a^*, H] = \sqrt{\frac{m\omega}{2}} [X - i \frac{P}{m\omega}, H]$

$$= \sqrt{\frac{m\omega}{2}} \left[\frac{\partial H}{\partial P} - \frac{i}{m\omega} \left(\frac{\partial H}{\partial X} \right) \right]$$

$$= \sqrt{\frac{m\omega}{2}} \left(\dot{X} - i \frac{\dot{P}}{m\omega} \right) = \sqrt{\frac{m\omega}{2}} \left(\frac{P}{m} - \frac{i}{\omega} X \right)$$

$$= \frac{\omega}{i} [X - i \frac{P}{m\omega}] = -i\omega a^*$$

例. 平面谐振子

$$H = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} m \omega_1^2 x^2 + \frac{1}{2} m \omega_2^2 y^2$$

$$\text{def } [J_x, J_y] = J_z$$

$$\text{def } H = \frac{1}{2m} (P_1^2 + P_2^2) + \frac{1}{2} m \omega_0^2 (q_1 + q_2)^2 + \frac{1}{2} m \omega_0^2 (q_1 - q_2)^2$$